Supplementary Images for Rebuttal

1 The loss curves for LLM experiments with more baselines and exact version

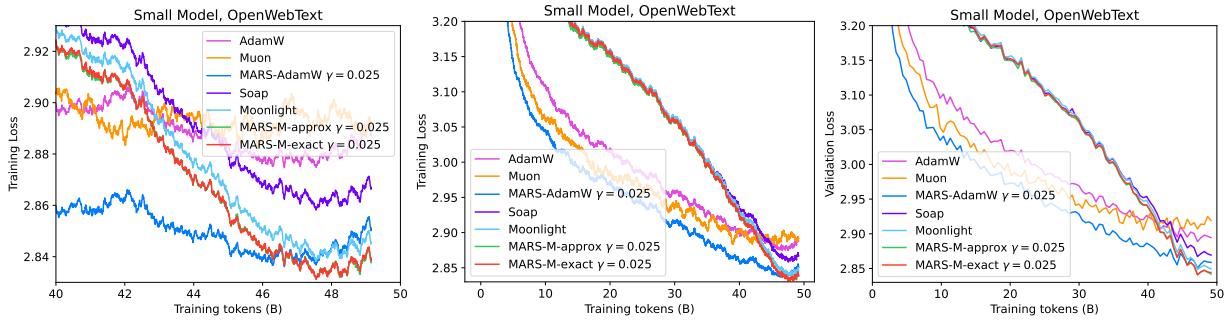

Figure 1: The training and validation loss curves of small-size (125M) models with different optimizers on the OpenWebText dataset. Here Muon is the original version proposed by (Keller Jordan, 2024) and Moonlight is the improved version with factor $0.2\sqrt{\max(m, n)}$ proposed by (Liu et al., 2025). The baseline curves of AdamW, Muon and MARS-AdamW are taken from (Yuan et al., 2025). **Left:** zoomed-in training loss curves. **Middle:** training loss curves. **Right:** Validation loss curves.

2 The dynamics of training

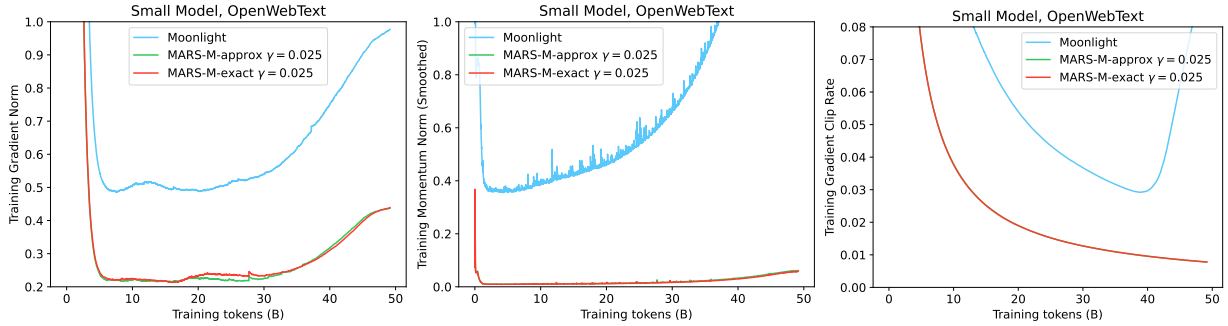

Figure 2: The dynamics during the training of small-size (125M) models with different optimizers on the OpenWebText dataset. The turning up in the final steps may be the result of decaying learning rate. **Left:** the gradient 2-norm. **Middle:** the smoothed training momentum norm. **Right:** the gradient clip rate (since we clip the original gradient to 1).

3 Additional results of Computer Vision tasks

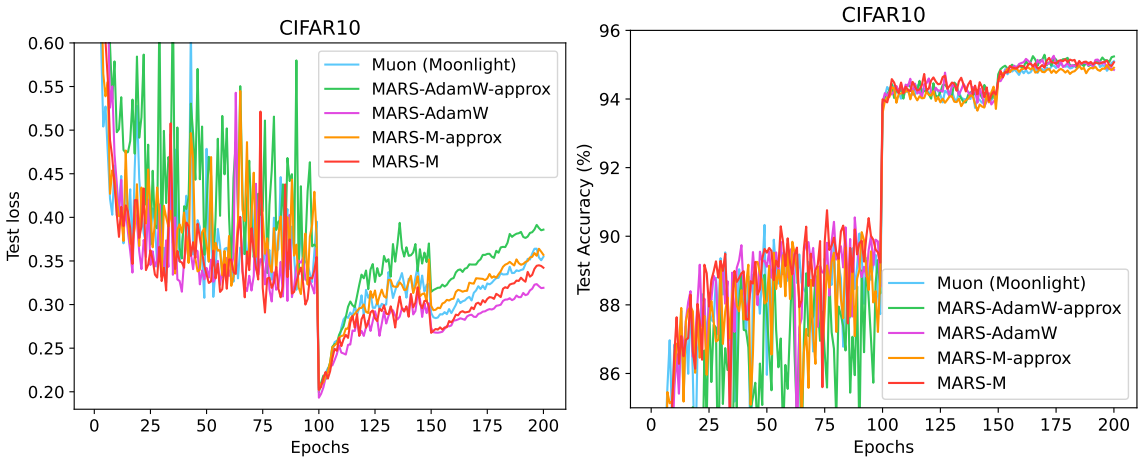

Figure 3: The test loss and test accuracy for different optimizers on CIFAR-10 dataset.

4 Ablation study on γ on medium-size GPT-2

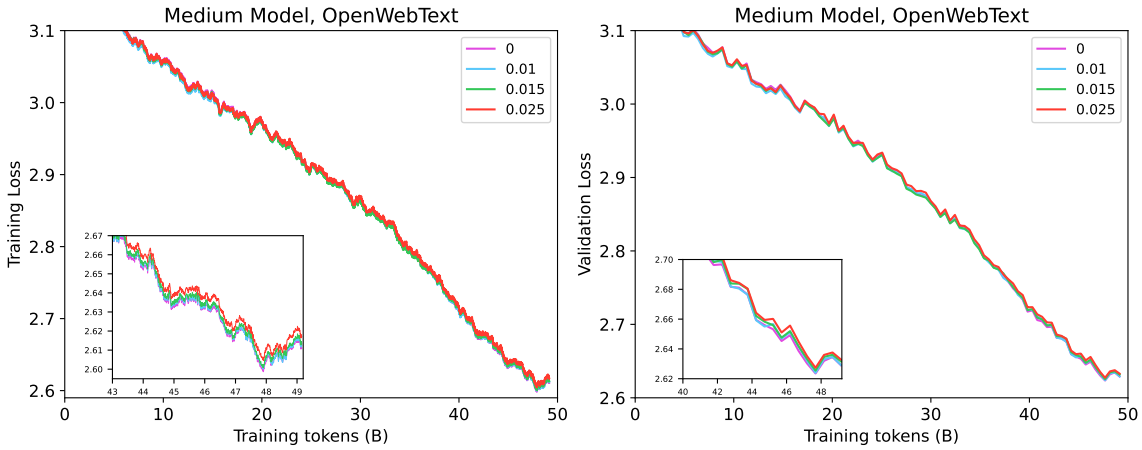

Figure 4: The zoomed-in training and validation loss of medium-size models (355M) trained with MARS-M with different γs on the OpenWebText 100B dataset.

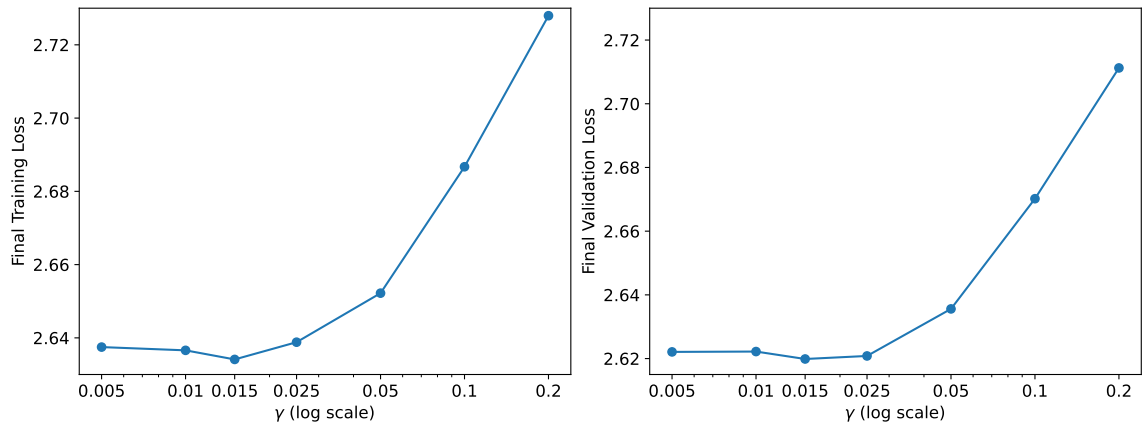

Figure 5: The final training and validation loss of medium-size models (355M) trained with MARS-M with different γs on the FineWeb-Edu 100B dataset.

5 Additional results of log-log gradient norm

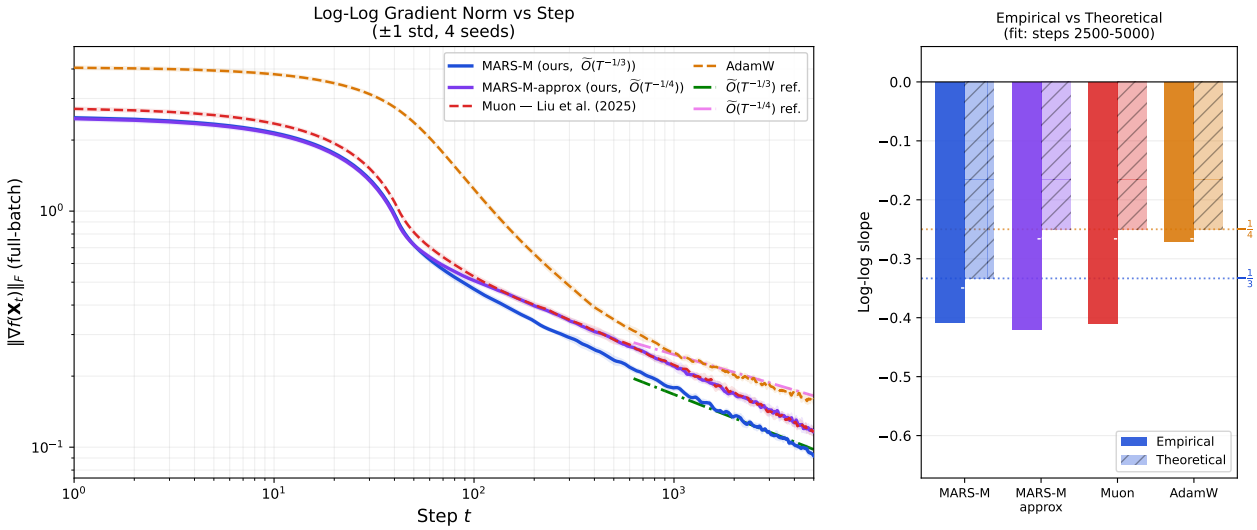

Figure 6: The dynamics of fitting $y = \tanh(WX)$ model with random Gaussian noise with a variance of 1 with different optimizers, where $W \in \mathbb{R}^{H\times d}, X \in \mathbb{R}^d, y \in \mathbb{R}^H$ with $H = 32, d = 64$ and a batch size of 64. Left: The log-log gradient norms w.r.t. step. We use 4 seeds and plot the $\mathcal{O}(T^{-1/4})$ and $\mathcal{O}(T^{-1/3})$ reference lines. Right: the slope of the curves on the log-log plot. Both the MARS-M and MARS-M-approx optimizers achieve a greater slope than the theoretical bound.